PrismMathQA: Agentic Math Tutoring with AVSD

Group 28

HUST

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Problem and Goal

Why math tutoring matters

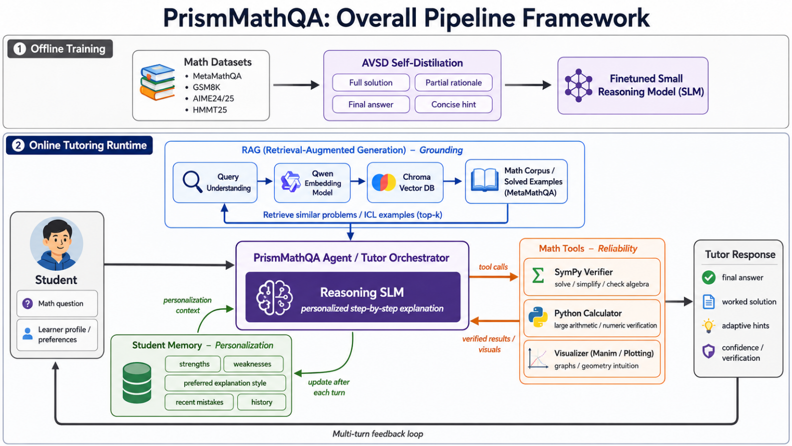
- Correctness matters: fluent explanations are not enough.
- Learning requires step-by-step guidance, not just final answers.
- Effective tutors adapt to learner context and follow-up questions.

PrismMathQA goal

Build a compact tutor that combines grounded examples, learner memory, exact symbolic verification, and a reasoning model trained for step-by-step mathematical behavior.

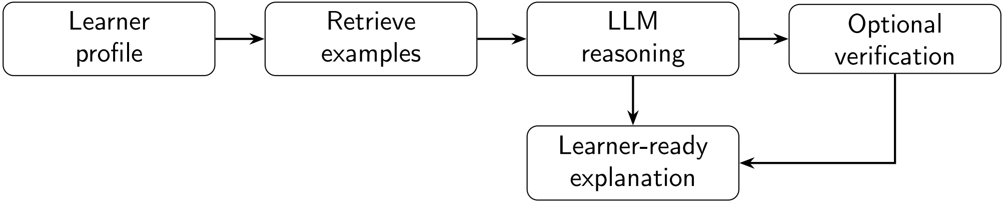
Core principle: the LLM coordinates specialized support instead of solving every subproblem alone.

Overall Framework



Offline: AVSD trains the reasoning model. **Online:** RAG, memory, and tools support each tutoring turn.

PrismMathQA Pipeline



Context

Grade level, preferred explanation style, weak areas, and recent conversation history guide the response.

Verification

The model asks for a SymPy-backed check when exact computation is more trustworthy than free-form generation.

Agentic Division of Labor

Reasoning LLM

Explains the solution, adapts tone and depth, and decides when external support is needed. It turns retrieved context and tool outputs into a coherent learner-ready explanation.

Learner memory

Stores profile fields, recent interactions, preferences, and weak areas for continuity across turns. This lets the tutor adjust pacing and avoid treating every question as a fresh session.

RAG over MetaMathQA

Uses Qwen3-Embedding-0.6B with Chroma to retrieve solved examples that expose relevant mathematical patterns and explanation styles. These examples ground the response in nearby problems instead of relying only on parametric memory.

SymPy verifier

Checks exact arithmetic, algebraic simplification, equation solving, and derivatives when reliability matters. The verifier handles deterministic computation that free-form generation may get wrong.

RAG Corpus: MetaMathQA Examples

Question type	Sample problem	Mathematical skill	Why it helps tutoring
Algebraic manipulation	Solve $987654321x - 123456789 = 555555555$ as a reduced fraction.	Linear equations, exact arithmetic, fraction reduction.	Shows clean symbolic transformations and exact intermediate steps.
Coordinate geometry	Find the perimeter form $a+b\sqrt{2}+c\sqrt{5}$ for a coordinate hexagon.	Distance formula, radicals, geometric modeling.	Translates coordinates or diagrams into algebraic quantities.
Functions and graphs	For $f(x) = x^2 - 4x + 1$, find the vertex, axis, and minimum value.	Completing the square, quadratic interpretation.	Connects algebraic manipulation to graph meaning.
Contest reasoning	Convert $(x, y) = (2\cos t - \sin t, 64\sin t)$ into $ax^2 + bxy + cy^2 = 1$.	Parameter elimination, trigonometric substitution.	Provides multi-step structure for hard derivations.

MetaMathQA is useful because retrieved items include both problem statements and worked solution text, so retrieval supplies a reasoning pattern rather than just an answer.

Why Fine-tuning Matters

Runtime support is necessary but not sufficient

RAG, memory, and tools improve grounding and reliability, but the base model still needs to produce coherent mathematical trajectories before external support is applied.

SFT

Dense traces, but static trajectories may not match the model's own rollout distribution.

RLVR / GRPO

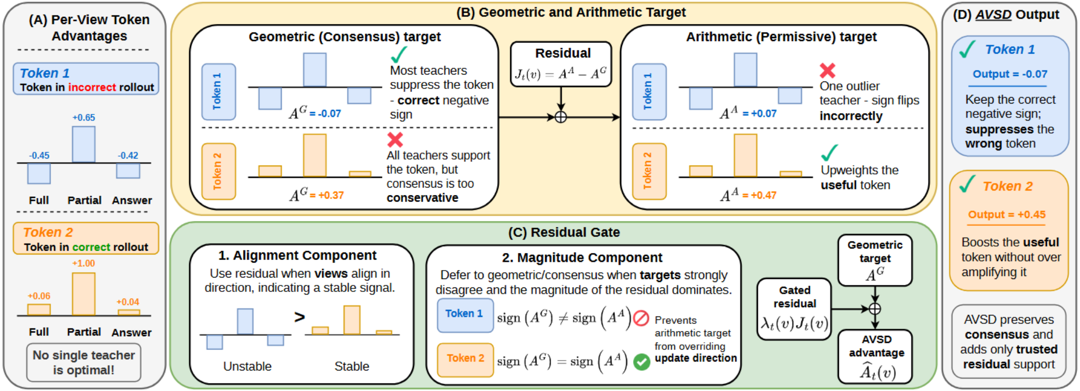
Verifiable rewards, but feedback is often sparse and final-answer oriented.

Self-distillation

Dense token-level learning signal on the model's own generated prefixes.

AVSD contribution: aggregate several privileged teacher views without trusting any single view unconditionally.

AVSD Core Idea



The student rolls out from the problem only; teacher mode evaluates the same prefixes under full solution, partial rationale, final answer, and concise hint.

Preparing Multi-view Data

View	Teacher context	Training role
Full solution	Complete worked derivation with final answer.	Provides dense reasoning supervision and exposes the full mathematical structure.
Partial rationale	Initial reasoning steps or a truncated rationale.	Guides the model toward a useful start without forcing every later step to match a reference.
Final answer	Gold answer without the full derivation.	Supplies compact correctness information while leaving room for the model's own explanation path.
Concise hint	Short hint such as a key substitution, theorem, or transformation.	Points toward a productive strategy without revealing the complete solution.

These four views are privileged training-time contexts; the deployed student model does not receive them at inference time.

Token-level Distillation Signal

Single-view reverse-KL intuition

At prefix $h_t = (x, y_{<t})$, compare the student distribution p_t with a privileged teacher distribution q_t:

$$A_t(v) = \log q_t(v) - \log p_t(v).$$

If $A_t(v) > 0$

The teacher assigns token v higher probability than the student, so training promotes that token.

If $A_t(v) < 0$

The teacher assigns token v lower probability than the student, so training suppresses that token.

Dense token feedback is valuable because many wrong math solutions fail at one local algebraic step rather than in the entire trajectory.

Consensus, Residual, and Gate

Consensus target

Geometric mean keeps tokens supported across views:

$$\tilde{q}_t^G(v) = \left(\prod_{m=1}^{M} q_t^{(m)}(v) \right)^{1/M}.$$

Arithmetic target and residual

Arithmetic mean captures view-specific union support:

$$q_t^A(v) = \frac{1}{M} \sum_{m=1}^{M} q_t^{(m)}(v),$$

$$J_t(v) = \log q_t^A(v) - \log \tilde{q}_t^G(v).$$

Adaptive gate

$$\widehat{A}_t(v) = A_t^G(v) + \lambda_t(v) J_t(v), \quad 0 \le \lambda_t(v) \le 1.$$

The gate admits residual evidence when views align and suppresses it when one privileged view dominates.

Reconstructed target

$$q_t^\star(v) \propto \tilde{q}_t^G(v) \exp(\lambda_t(v) J_t(v)).$$

Training Procedure and Lambda Gate

Training loop

1. Sample a math problem and privileged artifacts from a MetaMathQA subset.
2. Generate an on-policy rollout from the problem only.
3. Build four teacher contexts: full, partial rationale, answer, hint.
4. Evaluate teacher distributions at student prefixes.
5. Compute q_t^G, q_t^A, J_t, λ_t, $q_t^\star$.
6. Minimize reverse-KL to the reconstructed target.

Alignment component

$$C_t(v) = \frac{|A_t^G(v)|}{\frac{1}{M}\sum_{m=1}^{M} |\Delta_t^{(m)}(v)| + \epsilon}$$

High when teacher views promote or suppress the token in the same direction.

Magnitude component

$$R_t(v) = \frac{|A_t^G(v)|}{|A_t^G(v)| + J_t(v) + \epsilon}, \quad \lambda_t(v) = C_t(v)R_t(v)$$

Low when residual support is too large relative to consensus.

Benchmark Analysis: Reported Scores

Qwen3-4B checkpoint		Reported Avg@4			
Method	Training signal	GSM8K	AIME24	AIME25	HMMT25
Base	None	94.95	72.5	64.16	39.16
SFT	Static traces	89.91	56.3	49.83	29.6
GRPO	Verifiable rewards	95.81	73.14	65.70	41.82
OPSD	Single privileged view	94.98	73.31	65.46	43.31
RLSD	RLVR + self-distillation	**96.89**	**77.82**	<u>67.98</u>	<u>43.59</u>
AVSD	Multi-context views	<u>96.77</u>	<u>76.20</u>	**68.29**	**44.25**

Best results are bolded; second-best results are underlined. RLSD is strongest on GSM8K and AIME24, while AVSD is strongest on AIME25 and HMMT25. The table keeps the comparison on a fixed Qwen3-4B backbone.

View Ablation Analysis

AVSD view setting	GSM8K	AIME24	AIME25	HMMT25
1 view: full solution	94.98	73.31	65.46	43.31
2 views: full + partial rationale	95.58	74.27	66.40	43.62
3 views: full + partial rationale + answer	96.17	75.24	67.35	43.94
4 views: full + partial rationale + answer + hint	96.77	76.20	68.29	44.25

Insight 1: More views help

Performance rises consistently from 1 to 4 views, showing that the teacher family adds useful signal beyond a single full-solution view.

Insight 2: Harder math benefits

The largest gains appear on AIME24 and AIME25, where diverse views help stabilize multi-step contest reasoning.

Demo, Conclusion, and Future Work

PrismMath Tutor

LaTeX-rendered math tutor demo with first-turn RAG, student memory, and calculator tool reporting.

Student ID

s01

Nickname

An

Grade

12

Weak areas

Geometry

Strong areas

Calculus

Preferred explaining style

step-by-step

Question

Example: Verify the exact solution of \(987654321x - 123456789 = 555555555\), and compute \(x\) as a simplified fraction.

Ask Tutor

Turn: 1. RAG retrieval runs only on turn 1 of this browser session.

Log: `D:\Github\NLP-Capstone-Project-20252\logs\s01_20260606_235307.log`

Model Answer

Step 1. Write the parametric equations

$$x = 2\cos t - \sin t, \qquad y = X\sin t.$$

We are given $X = 64$, but we'll keep X symbolic for a moment and substitute the value at the end.

Step 2. Express $\sin t$ and $\cos t$ in terms of x and y

From the second equation

$$\sin t = \frac{y}{X}.$$

Insert this into the first equation:

$$x = 2\cos t - \frac{y}{X} \implies 2\cos t = x + \frac{y}{X} \implies \cos t = \frac{x + \frac{y}{X}}{2}.$$

So

$$\boxed{\sin t = \frac{y}{X}}, \qquad \boxed{\cos t = \frac{x + \frac{y}{X}}{2}}.$$

Step 3. Use the identity $\sin^2 t + \cos^2 t = 1$

$$\left(\frac{x + \frac{y}{X}}{2}\right)^2 + \left(\frac{y}{X}\right)^2 = 1.$$

Expand the first term:

$$\frac{(x + \frac{y}{X})^2}{4} + \frac{y^2}{X^2} = 1.$$